WT

Ch-1 Intro to web tech

1.1 Basics of World Wide Web (WWW)

- System of interlinked hypertext documents accessed via internet using web browsers

- Key components

1) client — Web browser (Firefox, Chrome)
2) Server — Stores & serves web pages
3) URL — Uniform Resource Locator — Address of a web resource
4) HTTP/HTTPS — Communication protocol
   Hyper Text Transfer Protocol Secure

- Example of URL

  https://www.example.com/index.html

1.2 HTTP Protocol — Stateless protocol used for communication between client & server.

☆ HTTP Methods

1) GET — Request data from server
2) POST — Send data to server
3) PUT — Update data
4) DELETE — Remove data
5) HEAD — Fetch headers only

- Example of GET Request : GET /index.html HTTP/1.0

1.3 ☆ HTTP Headers — Headers carry Metadata

- Request Headers
  - Host
  - User-Agent
  - Accept
  - Authorization
- Response Headers
  - Content Type
  - Content Length
  - Cache-Control
  - Set-cookie

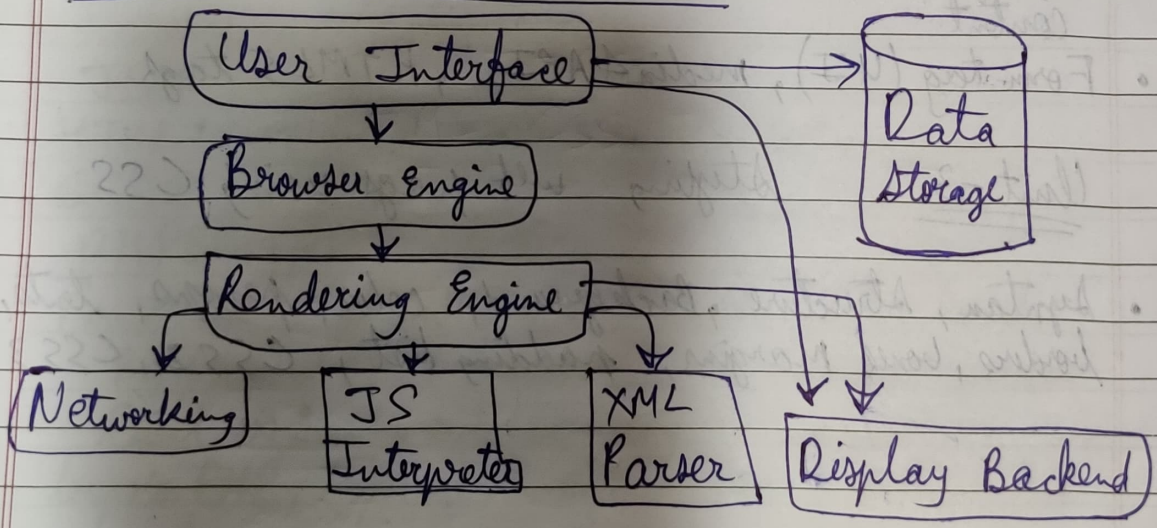
1.4 HTTP Request response structure

- HTTP Req → Method + URL + version
- Headers
- Body

HTTP Response → Status Line (200 OK)
Headers
Body (HTML data)

- Status codes
  - 200 – OK
  - 404 – Not found
  - 500 – Server Error

1.5 Web Browser Architecture



→ A Web Browser is a complex software system that takes a URL, fetches resources from internet, executes code and renders a visual web page. Internally it is divided into well-defined components.

1) User Interface (UI) — Represents the visible part. What you see & interact with

- Address Bar (URL Bar)
- Navigation Buttons (Back / Forward / Refresh)
- Tabs, Bookmarks, menus

It captures user actions (URL Entry, clicks, scrolls) and forwards them to the browser engine.

2) Browser Engine

- It acts as a bridge between UI & rendering engine
- It decides what to load & when

When user enters a URL, the browser engine:

1. Validates it
2. Requests data via networking layer
3. Passes received content to rendering engine.

3) Rendering Engine — Responsible for interpreting & displaying web content.

Converts raw web content into pixels on the screen.

- HTML parsing → creates DOM (Document Object Model)
- CSS parsing → creates CSSDOM
- DOM + CSSDOM → Render Tree
- Layout → calculates size & position
- painting → draws pixels on the screen.

Examples: Blink (Chrome), Gecko (Firefox), Web Kit (Safari).

4) Javascript Engine

- Executes client side scripts written in JS.
- It handles:
  - DOM Manipulation
  - Event Handling
  - AJAX / Fetch calls
  - Animations & interactivity
- JS can modify the DOM, forcing the rendering engine to re-layout or repaint the page.

5) Networking Component — Internet communication layer

- Handles HTTP/HTTPS requests & responses
- Downloads: HTML, CSS, JS
- Manages SSL/TLS for secure connection

Without this, the browser is just a UI shell.

6) Data Storage

Local Persistance

- Cookies
- Cache
- Local Storage / Session Storage
- Indexed DB
- Browser History

It improves performance (cache), maintain sessions (cookies) and stores user preferences.

1.6 Web Server Installation & Configuration

A web server handles HTTP requests.
Examples: Apache, Nginx, IIS.

॥ ॐ नमो भगवते वासुदेवाय नमः ॥

2 March 2026 - Monday - WT - EA - 705 - Moksha Patel Ma'am

PHP Old name Personal Home Page

PHP

Laravel → CRM Website
Customer Relationship Management

Wordpress

React JS & Node JS → JS + PHP (Server)

☆ PHP is server side scripting language means code runs on the server not in the user browser
Apache server runs PHP programs

UI Xampp, Wampp, Lampp — Provides Apache server & DB

C/ xampp / htdocs / demo / pr1.php

Print same line
echo below line

② Xampp Control Panel

Apache → Start — 80 port number

MySQL → Start for DB. — 3307

Pr1.php

PHP Code

```
<? php  → Starting Syntax
{
        ?7 → Ending Syntax.
```

☆ To print — use Echo "hello world";

```
<? php
    Echo "Hello World";
?7
```

localhost/demo/pr1.php

localhost:8080/demo/pr1.php

☆

Client Side Scripting	Server Side Scripting

JS var let const

classmate Date Page

PHP variables

To store temporary data $

① Start with dollar sign

② Case sensitive

```
<?php
    $xyz = 15;
    Echo $XYZ;   Wrong
    Echo $xyz;   ?>
```

No type declaration required. no need to mention

Data Types : String, integer, float, boolean, ~~Char~~ Array, object, null

Decision-Making statements & loops

- if - elseif - else
- switch
- for loop
- while loop
- do-while loop
- ✱ for-each loop

```
<?php
    $age = 37;
    if ($age >= 18) {
        Echo "Adult"; }
    else {
        Echo "minor"; }
    ?>
```

Q16) == vs === in JS?

Q17) Local Storage & Session Storage

~~Q18) map() & filter() methods in JS?~~

Q1) What are CSS Animations? Properties. Explain 3. — 3M

Q2) What is HTTP? How does it work? — 3M

Q3) Define: 1) WWW 2) Website 3) SOAP — 3M

Q4) Margin padding in CSS. Enlist Margin Properties — 3M

Q5) Wild Card Selectors (, ^, $) — 3M

*Q6) Document Object Model in JS with diagram — 3/6M

Q7) Write about any 2 mouse events, keyboard events & form events in JS. — 6M

Q8) Pop up boxes in JS — 3M

*Q9) Design Registration form in HTML — 6M

Q10) JSON — 3M

*Q11) Explain HTML tags: <select>, <input>, , <frameset>. — 3M

Q12) PHP vs HTML — 3M

*Q13) Explain as example about conditional statements of PHP — 6M

Q14) JS setTimeout() & setInterval().

*Q15) JS program on CRUD operations on JSON object — 6M

Q19) External CSS, Inline CSS, Internal CSS? — 3M

Q20) Client Side Scripting & Server Side Scripting — 3M

Q21) Loops in PHP? Explain types of loops in PHP — 6M

Q1) What are CSS Animations? Enlist properties. List any 3.

Ans1) CSS Animations are used to animate (add motion to) HTML elements without using Java Script. They allow smooth transitions between different styles over time using keyframes.

CSS animations are part of CSS3.

* Common Properties

1) animation-name: Specifies the name of the keyframe animation to be applied.

2) animation-duration: Defines how long the animation takes to complete eg 2s, 5s.

3) animation-delay: Specifies the delay before animation starts.

4) animation-iteration-count: Defines how many times the animation should repeat.

5) animation-direction: Specifies whether the animation should play forward, reverse or alternate.

Q2) What is HTTP? How does it work?

Ans 2) Hyper Text Transfer Protocol (HTTP) is a communication protocol used to transfer data between a web browser & web server on the internet.

It is the foundation of data communication in the World Wide Web.

* How HTTP works:

1) Client Request: The browser sends an HTTP request to the server eg GET, POST.

2) Server Processing: The Server recieves the request, processes it & prepares a response.

3) Server Reponse: The Server sends back an HTTP response containing status code (eg 200 OK) and requested data (HTML, CSS).

- HTTP is a communication protocol used for communication between client & server on the web.
- It works on a request response model where client sends request & server responds with data.

Q3) Define : 1) WWW 2) Website 3) SOAP

1) WWW : It is a system of interlinked web pages & resources accessed over the internet using a web browser. It uses HTTP protocol to retrieve & display content like text, images & videos.

2) Website : A website is a collection of related web pages stored on a web server & identified by a common domain name. It can contain text, images, videos & interactive elements.

3) SOAP (Simple Object Access Protocol) : SOAP is a protocol used for exchanging structured information between applications over the internet. It uses XML format for sending messages & works mainly over HTTP.

Q4) Margin & Padding in CSS. Enlist properties. — 3M

Ans4) Margin is the space outside the border of an element. It controls the distance between the element & other elements.

Padding is the space inside the border of an element. It controls the distance between the content & the border.

margin - top	padding - top
margin - left	padding - left
margin - bottom	padding - bottom
margin - right	padding - right
margin	padding.